

AI-POWERED RESUME SCREENING AND JOB CATEGORY PREDICTION SYSTEM

Submitted By

Name: Diksha Verma

Roll No.: 202401100800074

Branch: Information Technology

Semester: 5th

College: KIET Group of Institutions, Ghaziabad

This is to certify that the project entitled "**AI-Powered Resume Screening and Job Category Prediction System**" has been successfully completed by **Diksha Verma** in partial fulfillment of the requirements for the Bachelor of Technology degree in Information Technology at KIET Group of Institutions, Ghaziabad.

ACKNOWLEDGEMENT

I express my sincere gratitude to my faculty members and KIET Group of Institutions for providing guidance and support throughout the development of this project. I also thank my friends and classmates for their valuable suggestions and encouragement. Finally, I would like to thank my parents for their continuous support and motivation.

ABSTRACT

Recruitment is one of the most important processes in any organization. Human Resource departments receive thousands of resumes for different job positions. Manually screening these resumes is time-consuming and requires significant effort.

The objective of this project is to develop an AI-powered Resume Screening System capable of automatically classifying resumes into different job categories using Machine Learning and Natural Language Processing (NLP). The system uses TF-IDF Vectorization to convert textual resumes into numerical features and employs a Linear Support Vector Classifier (LinearSVC) for classification. A Flask-based backend API interacts with an HTML, CSS, and JavaScript frontend to provide a complete web application.

The developed system reduces manual effort, improves recruitment efficiency, and demonstrates the practical implementation of Artificial Intelligence in Human Resource Management.

Keywords: Artificial Intelligence, Resume Screening, Machine Learning, NLP, Flask, TF-IDF, LinearSVC.

TABLE OF CONTENTS

1. Introduction
 2. Problem Statement
 3. Objectives
 4. Literature Survey
 5. Existing System
 6. Proposed System
 7. System Architecture
 8. Dataset Description
 9. Data Preprocessing
 10. Machine Learning Model
 11. Technology Stack
 12. System Implementation
 13. API Documentation
 14. Results
 15. Advantages
 16. Limitations
 17. Future Scope
 18. Conclusion
 19. References
-

CHAPTER 1: INTRODUCTION

Artificial Intelligence has significantly transformed the recruitment process by automating repetitive tasks. Resume screening is the first step in recruitment, where recruiters manually analyze candidate resumes to determine their suitability for various job roles. Manual screening becomes difficult when thousands of resumes are received.

This project proposes an AI-powered Resume Screening System that automatically predicts the category of a resume using machine learning techniques. The application accepts resume text as input and returns the predicted job category, thereby reducing manual effort and increasing efficiency.

CHAPTER 2: PROBLEM STATEMENT

Organizations receive a large number of resumes for every job opening. HR professionals spend considerable time reading and categorizing resumes. This process is time-consuming, expensive, and prone to human error.

The objective of this project is to automate resume classification using Artificial Intelligence, enabling recruiters to quickly identify resumes belonging to different job domains.

CHAPTER 3: OBJECTIVES

The primary objectives are:

- Develop an AI-powered resume classification system.
 - Perform resume preprocessing using NLP techniques.
 - Convert textual data into numerical features using TF-IDF.
 - Train a Machine Learning model for classification.
 - Develop a Flask backend API.
 - Design an interactive frontend using HTML, CSS, and JavaScript.
 - Integrate frontend and backend into a complete web application.
-

CHAPTER 4: LITERATURE SURVEY

Several organizations have adopted Artificial Intelligence for recruitment automation. Research studies demonstrate that Natural Language Processing combined with Machine Learning significantly improves resume screening accuracy.

Traditional recruitment systems rely heavily on manual resume analysis, whereas modern AI systems use algorithms to classify resumes, rank candidates, and recommend suitable job profiles.

TF-IDF is widely used for representing textual information, while Support Vector Machines have consistently achieved high performance in text classification tasks.

CHAPTER 5: EXISTING SYSTEM

The existing recruitment process involves manual resume screening by HR professionals.

Disadvantages

- Time-consuming
 - Human errors
 - Inconsistent decisions
 - High operational cost
 - Difficult to process large volumes of resumes
-

CHAPTER 6: PROPOSED SYSTEM

The proposed system automatically predicts the job category of a resume using Machine Learning.

Features

- Resume classification
 - Fast prediction
 - User-friendly interface
 - Web-based application
 - AI-powered decision support
-

CHAPTER 7: SYSTEM ARCHITECTURE

User



Frontend (HTML + CSS + JavaScript)



Flask Backend API



TF-IDF Vectorizer



LinearSVC Model



Predicted Category



Display Result

CHAPTER 8: DATASET DESCRIPTION

The project uses the Resume Dataset downloaded from Kaggle.

Dataset Information

- Number of resumes: **2484**
- Columns:
 - ID
 - Resume_str
 - Resume_html
 - Category

The Resume_str column contains resume text used for training, while the Category column contains the corresponding job category.

CHAPTER 9: DATA PREPROCESSING

The preprocessing steps include:

- Removing special characters
 - Removing URLs
 - Removing punctuation
 - Removing numbers
 - Removing extra spaces
 - Lowercase conversion
 - TF-IDF Vectorization
 - Label Encoding
-

CHAPTER 10: MACHINE LEARNING MODEL

The following model was used:

Linear Support Vector Classifier (LinearSVC)

Reasons for choosing LinearSVC:

- High accuracy for text classification
- Fast training
- Efficient for high-dimensional TF-IDF features
- Suitable for multiclass classification

The trained model was saved using Joblib.

CHAPTER 11: TECHNOLOGY STACK

Programming Language

- Python

Machine Learning Libraries

- Pandas
- NumPy
- Scikit-learn
- Joblib

Backend

- Flask
- Flask-CORS

Frontend

- HTML
- CSS
- JavaScript

IDE

- Google Colab
 - Visual Studio Code
-

CHAPTER 12: SYSTEM IMPLEMENTATION

The project implementation consists of:

Model Training

- Loading dataset
- Data cleaning
- TF-IDF vectorization
- Model training
- Saving model

Backend Development

The Flask backend loads:

- resume_classifier.pkl
- tfidf_vectorizer.pkl
- label_encoder.pkl

The backend exposes the following API:

POST /predict

Frontend Development

The frontend consists of:

- Resume text area
 - Predict button
 - Prediction display
-

CHAPTER 13: API DOCUMENTATION

Endpoint

POST /predict

Request

```
{  
  "resume": "Resume Text"  
}
```

Response

```
{  
  "prediction": "HR"  
}
```

CHAPTER 14: RESULTS

The developed system successfully predicts the category of resumes.

Sample Predictions

- HR
- Data Science
- Java Developer
- Python Developer
- Testing
- DevOps

The application successfully integrates Machine Learning with a web interface.

(Insert the screenshots you captured here: homepage, resume input, prediction result, Flask terminal, project structure.)

CHAPTER 15: ADVANTAGES

- Fast prediction
 - Reduced manual effort
 - Easy to use
 - Scalable
 - Accurate classification
 - Web-based application
-

CHAPTER 16: LIMITATIONS

- Works only with trained categories.
 - Prediction quality depends on dataset quality.
 - Currently accepts pasted resume text instead of uploading PDF/DOCX files.
 - Does not rank resumes by suitability.
-

CHAPTER 17: FUTURE SCOPE

Future enhancements include:

- PDF/DOCX resume upload
 - Resume ranking
 - Skill extraction
 - Job recommendation
 - Candidate matching
 - Cloud deployment
 - Multilingual resume support
 - Integration with recruitment portals
-

CHAPTER 18: CONCLUSION

The AI-Powered Resume Screening and Job Category Prediction System successfully demonstrates the practical use of Artificial Intelligence in recruitment. By integrating TF-IDF vectorization, LinearSVC classification, Flask APIs, and a responsive web interface, the system automates the resume categorization process. The project reduces manual effort, improves efficiency, and provides a scalable solution for modern recruitment systems.

REFERENCES

1. Scikit-learn Documentation
2. Flask Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Kaggle Resume Dataset
6. Python Official Documentation